Week 3 Exercises

Exercise 1 Mean v. median

Suppose y follows a normal model with mean μ and SD σ. Then $\text{avg}(y)$ is the *best* unbiased estimator of μ. But $\text{median}(y)$ is also an unbiased estimator of μ. Use a Monte Carlo simulation to show how $\text{avg}(y)$ is a better estimator of μ than $\text{median}(y)$.

Exercise 2 Asymptotics as approximations

For a large number of repetitions, do the following:

1. Simulate a data set y with $N = 50$ observations from the exponential distribution with $\lambda = 2$.
2. Compute the point estimate $\hat{\lambda} = \frac{1}{\text{avg}(y)}$. Store this point estimate.

Compare the mean and SD of the simulated point estimates to the theoretical mean and SD of the *asymptotic* sampling distribution. We care, of course, about the *actual* sampling distribution, but we have to use an *asymptotic approximation* in practice. Discuss the differences and whether they seem important or unimportant.

Hint: For the exponential model, the asymptotic sampling distribution of $\hat{\lambda}$ is $\hat{\lambda} \overset{a}{\sim} N\left(\lambda,\ \frac{\lambda^2}{N}\right)$.

Optional: Go beyond the mean and SD. Use a table of quantiles of the simulated point estimates and theoretical distribution and/or plots of the CDFs to compare the simulated (i.e., actual) and theoretical distributions.

Exercise 3 German Tank Problem

Suppose a discrete uniform distribution from 0 to K. The pmf is $f(x; K) = \frac{1}{K+1}, \quad x \in \{0, 1, \dots, K\}$. The ML estimator is $\widehat{K} = \max(y)$. The method of moments estimator is $\widehat{K} = 2 \times \text{avg}(y)$. Use a Monte Carlo simulation to evaluate bias and variance of each estimator for a small ($N = 3$), medium ($N = 25$), and large sample size ($N = 1,000$). Which estimator do you recommend?

Note: This model violates several of the regularity conditions, so the usual asymptotic guarantees do not hold.

Exercise 4 Jensen's Inequality

Suppose the usual exponential model. The ML estimate of the rate λ is $\hat{\lambda} = \frac{1}{\text{avg}(y)}$. Using the invariance property, the ML estimate of the mean $\frac{1}{\lambda}$ is $\hat{\mu} = \frac{1}{\hat{\lambda}}$. Show analytically that the ML estimate of the mean is unbiased, while the ML estimate of the rate is biased.

Jensen's Inequality. If g is a convex function and X is a random variable with $\mathbb{E}[X]$ finite, then $\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])$, with strict inequality unless X is degenerate (i.e., constant almost surely).

Hint: First show that $\hat{\mu}$ is unbiased. Then use Jensen's inequality to show that $\hat{\lambda}$ is biased upward. You do not need to find the size of the bias, just use Jensen's inequality to show that the ML estimate of the rate is biased. To use Jensen's inequality, let $g(x) = 1/x$, and notice that $g(x)$ is convex on $(0, \infty)$ (since $g''(x) = 2/x^3 > 0$).

Exercise 5 Poisson

Suppose a Poisson model. The ML estimate of the mean parameter λ is $\hat{\lambda} = \text{avg}(y)$. Find an estimate of the SE using the observed Fisher information.

Exercise 6 `operations`

The code below loads the number of enforcement operations for each district in Bogota from Holland (2015) from the {crdata} package. See `?crdata::holland2015` for details.

```r
# load holland2015 data set from crdata package
holland2015 <- crdata::holland2015

# get operations in bogota only
ops <- holland2015$operations[holland2015$city == "bogota"]

# print
ops
```

```
 [1] 15 10 15 20  8  4  8 16  4 28  2  8  4  4  4  4  8  3  4
```

A Poisson model.

Task: Using the Poisson estimators from the previous question, find the ML estimate and SE estimate of λ for a Poisson model.

But before moving on, compare the SD of the predictive distribution (no need to simulate, it depends only and directly on λ) and the SD of the data. Why might this be a problem for the SE estimate?

A negative-binomial model.

For the Poisson model, the mean and variance of the outcome are directly linked. In fact, both equal λ. For most outcomes (almost all?), the variance is *much* larger than the average. This means a huge mismatch between the variance of the outcomes and the model.

The negative binomial model has an additional parameter r that allows the variance to be larger than (but not smaller than) the mean. For almost all datasets, this is a good default.

The negative binomial distribution is often parameterized by a mean $\mu > 0$ and a dispersion (or "size") parameter $r > 0$.[1] Its probability mass function is

$$\Pr(Y = y) \;=\; \frac{\Gamma(y+r)}{\Gamma(r)\, y!} \left(\frac{r}{r+\mu}\right)^{r} \left(\frac{\mu}{r+\mu}\right)^{y}, \qquad y = 0, 1, 2, \dots,$$

which has mean μ and variance $\mu + \mu^2/r$. In R, the base function `dnbinom()` directly supports this mean–size parameterization when the `mu` argument is supplied (e.g., `dnbinom(y, size = r, mu = mu)`).

Task. Use `optim()` to estimate the mean and its SE for the `ops` variable. Compare to the Poisson estimates.

[1]There are other common parameterizations that are less interpretable for our purposes.

Exercise 7 Exponential

Suppose an exponential model of y. We know that the ML estimate of the rate λ is $\hat{\lambda} = \frac{1}{\text{avg}(y)}$. By the invariance property, the ML estimate of the mean $\mu = \frac{1}{\lambda}$ is $\hat{\mu} = \frac{1}{\hat{\lambda}}$.

1. Use the observed Fisher information to find a closed-form estimate of the SE of $\hat{\lambda}$.
2. Then use the delta method to find a closed-form estimate of the SE of $\hat{\mu}$.

Hint: For an exponential model with rate λ, the density is $f(y; \lambda) = \lambda e^{-\lambda y}$, so $\ell(\lambda) = \sum \big(\log \lambda - \lambda y_i \big)$.

Exercise 8 Delta method, by hand

Suppose you obtain the ML estimate $\hat{\theta} = (\hat{a}, \hat{b}) = (2, 1)$ with an estimated covariance matrix of

$$\widehat{\text{Var}}(\hat{\theta}) = \widehat{\text{Var}}(\hat{a}, \hat{b}) = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix}.$$

However, you want to estimate the quantity of interest $q = \tau(a, b) = \dfrac{a}{b}$. Find $\hat{q}$ using the invariance property and estimate the SE of $\hat{q}$ using the delta method. (Find the gradient analytically.) Confirm your work with R.

Exercise 9 (α, β) to SD

In the notes, we saw how to use the invariance principle and the delta method to obtain an ML estimate of the mean μ and its SE from ML estimates of α and β and their estimated covariance matrix.

Similarly, use the invariance principle and the delta method to find an ML estimate of the SD σ and its SE. Recall that $\sigma = \sqrt{\dfrac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}}$ for the beta distribution. Use a numerical gradient.

Exercise 10 Turnout rates

Load a tibble version of the 2024 turnout data from the UF Election Lab into R.

Model the proportion of the voting-eligible population that voted using a beta distribution. Estimate the parameters of the beta distribution α and β as well as the mean $\mu = \frac{\alpha}{\alpha+\beta}$ and the SD $\sigma = \sqrt{\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}}$ }$. Use a parametric bootstrap to estimate the SE of each.

Exercise 11 `duration`

Part 1. *Parametric models.*

Model `duration` in the `coalition` data set (from the **brglm2** package) using three different distributions: an exponential distribution, a Weibull distribution, and a log-normal distribution.

For each distribution:

1. Estimate the parameter(s) using maximum likelihood.
2. Estimate the covariance matrix.
3. Estimate the mean using the invariance property.
4. Estimate the SE using the delta method.

Hint: Use `optim()` and `numDeriv::grad()`. Complete each step for the exponential model, then copy-and-paste the code and make the needed changes for the Weibull and log-normal.

Part 2. Classical estimate.

Compute the classical mean and SE estimates by taking the average $\hat{\mu} = \text{avg}(y)$ and the standard error $\widehat{\text{SE}}(\hat{\mu}) = \dfrac{\text{SD}(y)}{\sqrt{N}}$.

Part 3.

Compare these four estimates and their standard error estimates. What stands out? Explain.

Some Background.

When we model duration data, we often focus on the hazard function, which describes the relative chance of an event at time t, given that the unit has survived until t. The hazard tells us how the risk of failure changes with time.

The *exponential distribution* is the simplest model. The density is

$$f(t \mid \lambda) = \lambda e^{-\lambda t}, \quad t > 0,\ \lambda > 0,$$

and the mean is $\mathbb{E}[T] = 1/\lambda$. The hazard is $h(t) = \lambda$, which is constant over time. This means the chance of failure in the next instant does not depend on how long the unit has lasted so far. This "memoryless" property makes the exponential convenient, but often unrealistic.

The *Weibull distribution* generalizes the exponential by allowing the hazard to change with time. The density is

$$f(t \mid k, \lambda) = \frac{k}{\lambda}\left(\frac{t}{\lambda}\right)^{k-1} \exp\left[-\left(\frac{t}{\lambda}\right)^{k}\right], \quad t > 0,\ k, \lambda > 0,$$

and the mean is $\mathbb{E}[T] = \lambda\,\Gamma(1 + 1/k)$. The hazard is

$$h(t) = \frac{k}{\lambda}\left(\frac{t}{\lambda}\right)^{k-1}.$$

When $k = 1$, this reduces to the exponential with a constant hazard. If $k > 1$, the hazard increases with time, which might capture processes like aging or wear-out. If $k < 1$, the hazard decreases with time, which might represent situations where early failures are more likely. This flexibility makes the Weibull especially useful.

The *lognormal distribution* assumes that the log of the durations is normally distributed. The density is

$$f(t \mid \mu, \sigma) = \frac{1}{t\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln t - \mu)^2}{2\sigma^2}\right), \quad t > 0,\ \sigma > 0,$$

and the mean is $\mathbb{E}[T] = \exp(\mu + \frac{1}{2}\sigma^2)$. The hazard is non-monotone: it starts low, rises for a while, and then falls. This "hump-shaped" hazard works well when events are most likely after some waiting time, but less likely if they haven't occurred by then—for example, the adoption of a new technology or the incubation period of a disease.

Exercise 12 Two CIs

Consider a simple Bernoulli model for $y \in \{0, 1\}$ with parameter $\pi \in [0, 1]$. Create a small sample with a rare event: take $N = 20$ trials with exactly one success.

Part 1. *Wald confidence interval.*

Compute the ML estimate $\hat{\pi} = \operatorname{avg}(y)$, the Wald SE $\widehat{\mathrm{SE}}(\hat{\pi}) = \sqrt{\hat{\pi}(1 - \hat{\pi})/N}$, and the usual 95% Wald confidence interval $\hat{\pi} \pm 1.96 \cdot \widehat{\mathrm{SE}}(\hat{\pi})$. What do you notice about the interval relative to the parameter space $[0, 1]$?

Part 2. *Parametric bootstrap confidence interval.*

Use a *parametric bootstrap* to obtain a 95% percentile confidence interval for π based on the fitted model $\text{Bernoulli}(\hat{\pi})$. Compare the bootstrap interval with the Wald interval. What stands out? Briefly explain.

Hint: For the bootstrap, repeatedly simulate $y_1^{*(b)}, \ldots, y_N^{*(b)} \overset{iid}{\sim} \text{Bernoulli}(\widehat{\pi})$, recompute $\widehat{\pi}^{*(b)} = \text{avg}(y^{*(b)})$, and then take the empirical 2.5^{th} and 97.5^{th} percentiles of $\{\widehat{\pi}^{*(b)}\}$.